Python Activity-Based Learning
Program 5: Multi-Plan Telecommunications Cost Optimization

Problem Statement

Compare mobile plans using monthly data usage, validity, and budget. Calculate excess data charges and identify the most economical plan that satisfies the user's needs.

Objectives

• Evaluate multiple mobile plans.

• Calculate extra charges when usage exceeds included data.

• Select the cheapest plan within budget and validity requirements.

Assumptions

• Each plan has a name, price, included data in GB, validity in days, and excess charge per GB.

• The user must have enough validity and data for the entered requirements.

• Any usage above included data is charged per GB; the lowest total cost within budget is selected.

Algorithm

1. Read required data, validity, and budget.

2. Read the number of plans.

3. For each plan, read its details.

4. Calculate excess data and total cost.

5. Check data, validity, and budget conditions.

6. Keep the eligible plan with the lowest total cost.

7. Display the selected plan or a no-plan message.

Flowchart

```
START → Read user requirements → Read plans → Calculate excess data and cost → Check validity/data/budget → El
```

Pseudocode

```
BEGIN
  READ required_data, required_days, budget, number_of_plans
  best_cost = very large number
  FOR each plan
    READ name, price, included_data, validity, extra_rate
    extra = maximum of 0 and (required_data - included_data)
    total = price + extra * extra_rate
    IF validity >= required_days AND total <= budget
       AND total < best_cost THEN save this plan
    END IF
  END FOR
  IF a plan was saved THEN DISPLAY it
  ELSE DISPLAY no suitable plan
END
```

Python Source Code

```
required_data = float(input("Required data in GB: "))
required_days = int(input("Required validity in days: "))
budget = float(input("Your budget: "))
n = int(input("Number of plans to compare: "))

best_plan = ""
best_cost = float("inf")
```

```
for i in range(n):
    print("\nPlan", i + 1)
    name = input("Plan name: ")
    price = float(input("Plan price: "))
    included_data = float(input("Included data (GB): "))
    validity = int(input("Validity (days): "))
    extra_rate = float(input("Extra charge per GB: "))

    extra_data = max(0, required_data - included_data)
    total_cost = price + extra_data * extra_rate

    print("Total cost:", round(total_cost, 2))

    if (validity >= required_days and
            included_data + extra_data >= required_data and
            total_cost <= budget and total_cost < best_cost):
        best_cost = total_cost
        best_plan = name

if best_plan == "":
    print("\nNo plan satisfies your requirements and budget.")
else:
    print("\nMost economical eligible plan:", best_plan)
    print("Total cost: Rs.", round(best_cost, 2))
```

Test Case

Requirements: 2 GB, 28 days, budget ■300. Plan A: ■199, 1 GB, 28 days, ■50/extra GB. Plan B: ■249, 2 GB, 28 days, ■50/extra GB.
Expected: Plan A total ■249; Plan B total ■249. The first lowest-cost plan (Plan A) is selected.

Sample Output

```
Plan 1 Total cost: 249.0
Plan 2 Total cost: 249.0
Most economical eligible plan: Plan A
Total cost: Rs. 249.0
```

Conclusion

The program compares mobile plans by calculating total cost and checking the user's data, validity, and budget requirements.